

Factors Affecting Final Grades

Introduction: Student academic performance is impacted by a combination of academic habits, demographics, behavioral choices, and environmental factors. Among these, alcohol consumption is often assumed to negatively affect students' grades and school outcomes. This is because alcohol consumption is associated with poor sleep, lower concentration, and less study time. However, the real impact of alcohol consumption on students' final grades leaves a lot to be discovered and is not straightforward. The project began with two research questions, the first one being: Does alcohol consumption significantly affect students' likelihood of passing their course? After some beginning analysis, we broadened our scope and began to also think: What factors can meaningfully predict students' success in a course? We first explored an ANOVA model for predicting numeric final grades, but the model explained very little variation (~ 1%) and violated key modeling assumptions. We also looked at a mixed effects logistic regression model with a random effect for subject (math and language), but this model also did not give much insightful information. We finally selected a binary logistic regression model to conduct our analysis. Specifically, we modeled the probability that a student passes a course from academic behaviors, demographic factors, and alcohol consumption using logistic regression. Final grades can show a clear separation between pass and fail, and we set the standard at 60%. Logistic regression is a valid choice for modeling this data when using a binary outcome because it provides interpretable odds ratios. The odds ratios allow us to quantify how different variables influence academic success.

Data Description & Exploratory Analysis: The data used in this project comes from Kaggle, where we found a dataset titled "Student Alcohol Consumption". This data was collected from a

survey conducted of 1044 students in two Portuguese secondary schools and includes detailed information on students' demographics, family background, academic behavior, alcohol use, and school performance, ages 15 to 22. The data contained 33 total variables between math and language subjects, but we focused on the final grade response variable (0 to 20 scale), and a subset of predictors: sex (Male/Female), studytime (time used to study ranked 1-5), failures (number of past academic failures), activities (extracurricular participation Yes/No), internet (internet access at home Yes/No), goout (time spent going out ranked 1-5), Dalc (weekday alcohol consumption ranked 1-5), Walc (weekend alcohol consumption ranked 1-5), and absences (number of absences). The dataset was already clean with no missing values; the only preprocessing step required was merging the math and language datasets, which shared identical variable structures. We began by running summary statistics on the clean and merged dataset that we labeled as *students_clean*.

Variable	Min	1st Qu.	Median	Mean	3rd Qu.	Max
age	15	16	17	16.73	18	22
studytime	1	1	2	1.97	2	4
absences	0	0	2	4.435	6	75
failures	0	0	0	0.264	0	3
goout	1	2	3	3.156	4	5
Dalc	1	1	1	1.494	2	5
Walc	1	1	2	2.284	3	5
final_grade	0	10	11	11.34	14	20

Table 1: Summary statistics for the variables used in the alcohol consumption dataset

Summary statistics showed that most students were 16–18 years old, with slightly more females than males. Alcohol consumption was lower than expected on average (mean Dalc = 1.49; mean

Walc = 2.28), and final grades centered around a mean of 11.34 (~57%). Our next analysis step was creating plots to explore relationships between variables. Scatterplots and boxplots of alcohol use vs. final grade showed no clear linear pattern, and most students reported low alcohol consumption (*Figure B1.1*). Weekend alcohol consumption showed a similarly weak relationship with final grades, with little separation across drinking levels (*Figure B1.2*). Time spent going out also showed overlap in final grades, suggesting only a small association (*Figure B1.3*). This motivated shifting our focus toward other predictors. Stronger patterns emerged when comparing academic behaviors to the final grade. Prior failures (*Figure B1.8*) and absences (*Figure B1.7*) both showed a strong negative relationship with final grade and passing probability. Study time demonstrated a positive association, with more study time linked to higher final grades and greater odds of passing (*Figure B1.6*). Access to the internet (*Figure B1.4*) and participation in extracurricular activities (*Figure B1.5*) both correlated with higher final grades. Overall, the EDA showed us that alcohol consumption may not be a strong predictor of academic success in this dataset, whereas measures of academic engagement could be much more important factors.

Model Building: Because our original research question focused only on alcohol consumption and academic performance, we first attempted a two-way ANOVA using sex, weekday alcohol consumption (Dalc), and their interaction to predict final grade. This model violated model assumptions, as residual diagnostics did not support normality. It only explained about 1% of the variation in final grades, and none of the predictors or interactions were statistically significant. We then tried a mixed effect model with a random intercept for subject, but the random intercept variance was extremely small, and it did not show meaningful subject-level variation. Finally, we created a binary pass/fail response variable and ran a binary logistic regression model. If the final

grade was less than 12, it was a fail, and 12 or greater was a pass because 12/20 is a 60% grade, and we got the most even amount of passes and failures. This cutoff aligns with the Portuguese grading system, where a score between 10 and 12 out of 20 represents a passing grade. There were 534 failures and 510 passes. Our full model included:

$$\text{logit}(P(\text{pass})) = \beta_0 + \beta_1 \text{sex} + \beta_2 \text{internet} + \beta_3 \text{failures} + \beta_4 \text{absences} + \beta_5 \text{activities} + \beta_6 \text{studytime} + \beta_7 \text{goout} + \beta_8 \text{Dalc} + \beta_9 \text{Walc}$$

We then converted the coefficients to odds ratios and used the findings from those to draw further conclusions. When looking at model diagnostics, the pseudo $R^2 = 0.135$, which is reasonable for observational behavioral data. While the pseudo- R^2 value of 0.135 may appear modest, values like this are common in observational behavioral studies and indicate predictive improvement over a null model. Cook's distance and leverage showed a small amount of highly influential observations driven largely by extreme absences. Residual deviance decreased, meaning an improvement over a model with no predictors. Overall, diagnostics suggested this logistic model was appropriate and reasonably stable.

After this model, we wanted to look at a reduced model that only includes a few significant variables. We first fit a reduced model using only the four strongest predictors identified in the full model: failures, absences, studytime, and internet access (*Figure B2.3*). This reduced model remained statistically significant, but actually had a higher AIC than our full logistic model with all the variables (*Figure B2.6, B3.1*). This indicated that some explanatory power was being lost when removing too many variables. A likelihood ratio test further confirmed that removing additional predictors led to a significantly worse fit. Because we found a small number of highly influential observations after completing Cook's plots, we also refit the model after excluding these points (*Figure B2.2*). After removing all the influential points using Cook's distance, we

found that the model had a lower AIC than any model we created before (*Figure B2.4, B3.2*).

Every single variable used in our model was also significant in this model, showing how our Walc and Dalc variables are highly impacted by outliers.

A logistic regression model showing how just subjects and alcohol consumption affect the final grade was made (*Figure B2.5*). However, the interaction term made with Dalc and the subject was not significant, so no further analysis was done.

The final two models we created were logistic regression models that only included females (*Figure B2.8*) and only males (*Figure B2.7*), respectively. These models used all of the variables from our logistic full model. For the male logistic model, the AIC went down to 537.38, but found only activities, failures, and Walc significant. The female model had a higher AIC at 736.78, and found internet, failures, absences, and study time significant. These models showed how academic variables can be similar across sexes, but also how different behavioral factors are between the sexes.

Overall, after making all of these models and keeping in mind our research questions, we decided to use our full logistic regression model as our final model.

Results: Our full logistic model identified multiple significant predictors of whether a student passed the course (≥ 12). The final model resulted in:

$$0.145 - 0.007sexmale + 0.582internetyes - 1.544failures - 0.039absences + 0.289activitiesyes + 0.227studytime - 0.144goout - 0.1817Dalc - 0.0349Walc$$

The table below shows the coefficients and statistics from the full logistic regression model.

	Estimate	Std. Error	Pr(> z)
Intercept	0.144512	0.309458	0.640511

SexMale	-0.007203	0.148722	0.961373
internetyes	0.582003	0.171085	0.000669
failures	-1.543851	0.206068	6.79e-14
absences	-0.039215	0.012659	0.001950
activitiesyes	0.288957	0.138443	0.036871
goout	-0.143722	0.064514	0.025896
studytime	0.227323	0.085835	0.008088
Dalc	-0.181690	0.101195	0.072583
Walc	-0.034900	0.074394	0.638979

Table 2: Coefficient estimates and model statistics for the full logistic regression model

Consistent with expectations, the alcohol consumption variables were both insignificant and were not strong predictors of final grade, and academic behaviors were the strongest predictors. Failures, absences, and studytime were highly significant; internet access and participation in extracurricular activities also were significant. Social behaviors such as going out had a small but significant negative effect. To improve interpretability, we found the odds ratios of each predictor from the full model.

term	odds_ratio	pct_change
(Intercept)	1.155	15.5%
sexMale	0.993	-0.7%
internetyes	1.790	79%
failures	0.214	-78.6%
absences	0.962	-3.8%
activitiesyes	1.335	33.5%
goout	0.866	-13.4%

studytime	1.255	25.5%
Dalc	0.834	-16.6%
Walc	0.966	-3.4%

Table 3: Odds ratios for predictors from full model

Looking at the most significant predictors, students with home internet had 79% higher odds of passing, and each additional prior failure reduced the odds of passing by ~80%. For example, holding all other variables constant, a student with one additional prior failure has approximately 1/5 the odds of passing compared to an otherwise similar student. The alcohol consumption predictors showed little to no effects. The fitted vs. actual boxplot showed clear separation in predicted probabilities between students who passed and those who failed.

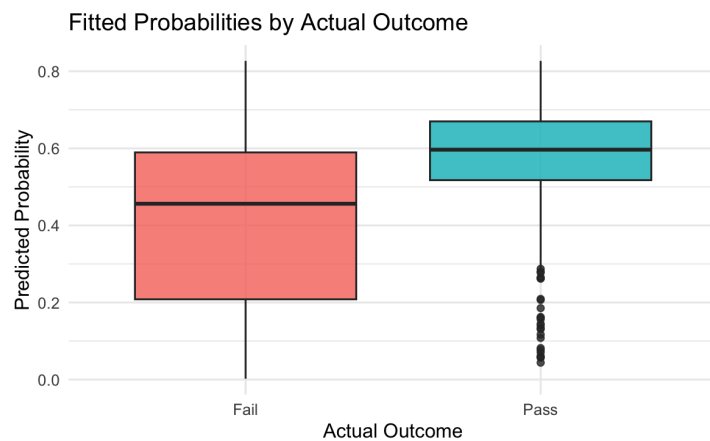


Figure 1: Boxplot showing fitted vs actual probabilities

This means that the model assigns higher predicted probabilities to actual pass outcomes. The prediction versus actual table showed that the model correctly classified a majority of both passing and failing students, with an overall accuracy of approximately 64% (*Figure B2.1*). We next compared the reduced model with the top four strongest predictors to the full model. All four predictors remained significant, but the reduced model had a higher AIC (1281.3) compared

to the full model (1271.3), which indicates a weaker fit (*Figure B3.1*). Next, we examined each gender separately by fitting different logistic regression models for males and females. These models included the same predictors as the full model but only looked at each gender's separate responses. (*Figures B2.7 and B2.8*). The male model identified failures, activities, and Walc as significant predictors, while the female model identified failures, internet access, study time, and absences. It is interesting how the male model shows the alcohol variable as significant, whereas it is not significant in the female model. However, because these models reduce sample size and do not answer our research questions, they were not selected as final models. Model diagnostics supported the validity of the full logistic regression model. In summary, our modeling results show a clear pattern, which is that students' academic habits and access to resources have a stronger influence on success than their reported alcohol use.

Discussion & Conclusion: The goal of this project was to determine whether alcohol use affects students' academic performance and to find which other behavioral or demographic factors best predict student success. Our results consistently showed that alcohol consumption is not a significant predictor of passing or failing a class because both Dalc and Walc were insignificant in the full logistic regression model. Even though the male-only model found a weak significance for weekend drinking, this did not hold for the female dataset or across the full dataset. Across every model we created, academic habits remained as the strongest predictors of success. Prior failures consistently had the largest effect, with each additional past failure reducing the odds of passing by almost 80%. Absences also lowered the likelihood of passing, showing that consistent attendance is a crucial component of academic success. Study time had a positive effect across all models, suggesting that even small increases in study habits can

improve student outcomes. Access to the internet and participation in extracurricular activities were also positively correlated with passing, which could reflect that greater resources or engagement with school can improve success. The reduced model helped confirm that failures, absences, study time, and internet access are the most influential predictors. However, the reduced model fits worse than the full model, meaning that explanatory power is lower when other behavioral variables are removed from the model. Our gender specific models identified that while the academic predictors remain significant across groups, the social and lifestyle variables may differ for male and female students. We faced some limitations throughout the completion of this project. First, the dataset was collected only from two schools in Portugal, which limits how the findings can be generalized to other populations or educational systems. Also, we created our own pass/fail variable (final grade ≥ 12), which introduces subjectivity and could influence model behavior. The data were self-reported, so alcohol use and behavioral variables could be underreported or have recall bias. Finally, we explored an ANOVA framework early in the analysis, but it explained only about 1% of the variance in final grades, reinforcing that logistic regression was more appropriate. Despite the initial focus on alcohol, our findings show that improving academic habits may have a greater impact on student performance. Schools and policymakers could benefit from programs that provide additional support to students with a history of academic failures, encourage structured study routines, reduce absences, and ensure internet access for all students. Future work could build on this project by using GPA instead of thresholds or using a larger and more diverse sample. Expanding the analysis to multiple ages or grades could improve the model, along with looking at additional variables such as mental health, sleep habits, or socioeconomic status. Since this dataset was only from Portugal grades 10-12, we could also explore other datasets that explore this age range

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in other countries. We could do comparison models of the countries and find the variables that overlap and differ across countries across the world. With a richer and more generalized dataset, mixed-effects models could be explored more. Overall, this study highlights that academic habits and access to resources outside of school are far more predictive of academic success than alcohol use alone.

Appendix:

A. [R Coding Script](#)

B. [Kaggle Dataset](#)

C. Supplementary Visualizations

B1. Exploratory Visualizations

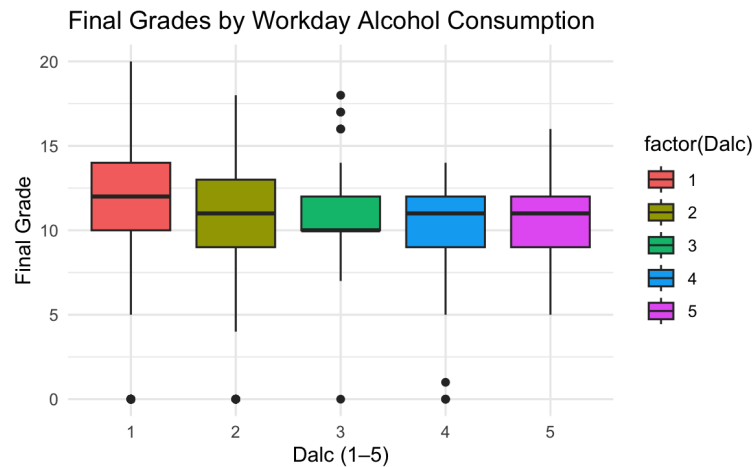


Figure B1.1 Final Grades by Workday Alcohol Consumption – Boxplot showing how workday alcohol consumption compares to final grades.

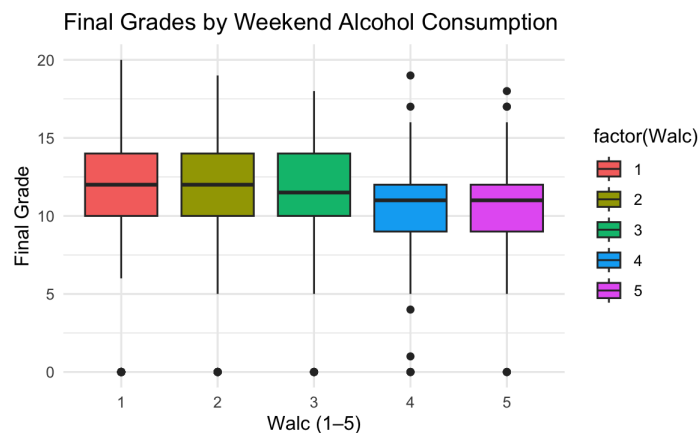


Figure B1.2 Final Grades by Weekend Alcohol Consumption – Boxplot showing how weekend alcohol consumption compares to final grades.

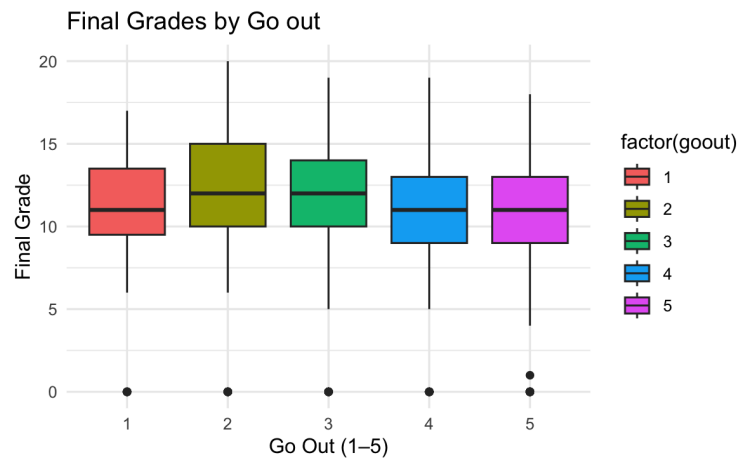


Figure B1.3 Final Grades by Going Out – Boxplot showing how going out compares to final grades.

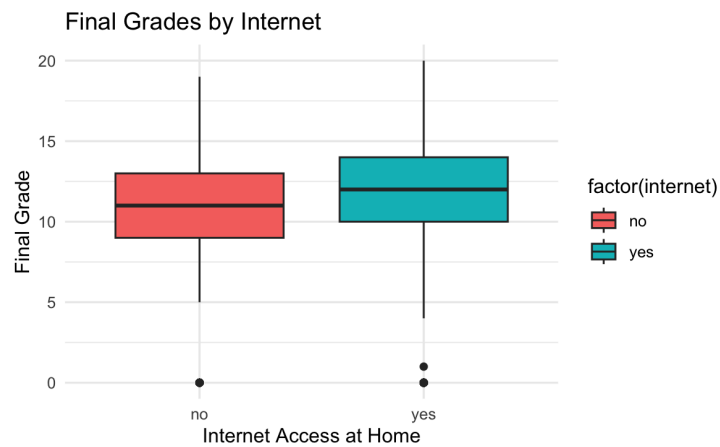


Figure B1.4 Final Grades by Internet Access – Boxplot showing how having access to the internet compares to final grades.

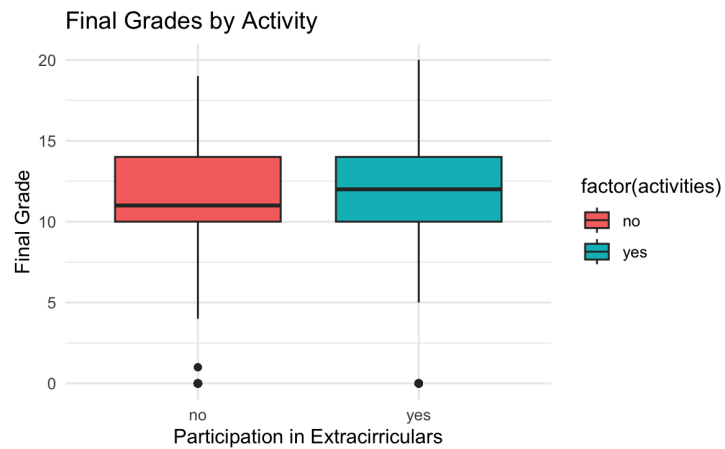


Figure B1.5 Final Grades by Participation in Extracurricular Activities – Boxplot showing how participation in extracurricular activities compares to final grades.

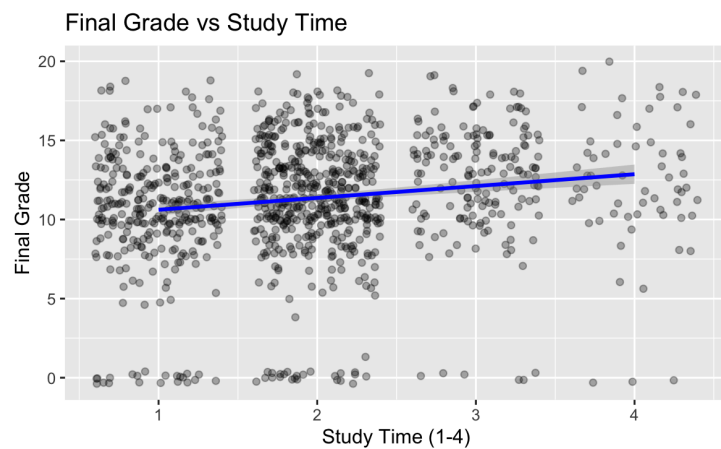


Figure B1.6 Final Grades vs. Study Time – Scatterplot showing how study time compares to final grades.

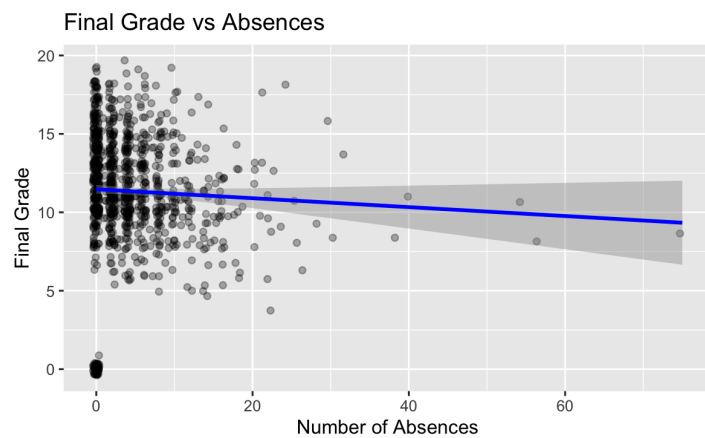


Figure B1.7 Final Grades vs. Absences – Scatterplot showing how the number of absences compares to final grades.

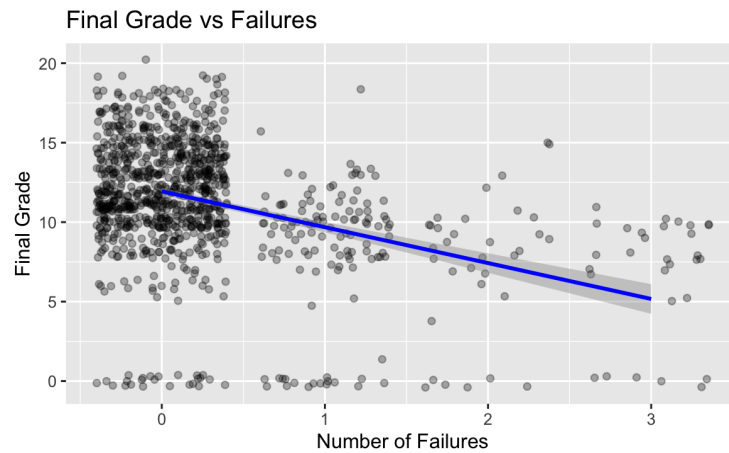


Figure B1.8 Final Grades vs Failures – Scatterplot showing how the number of past failures compares to final grades.

B2. Model Diagnostics

	Actual	
Predicted	Fail	Pass
Fail	414	259
Pass	120	251

Figure B2.1 Prediction vs. Actual table – predicted values versus actual values for the pass/fail response variable

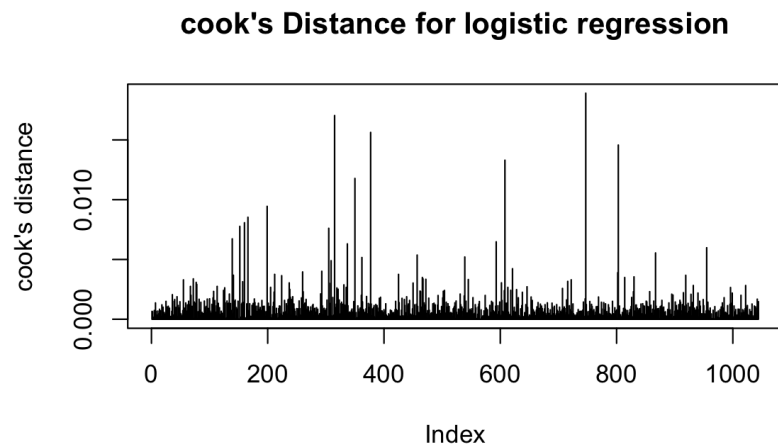


Figure B2.2 Cook's distance – Cook's distance influential points for the full logistic regression model.

	Estimate	Std. Error	Pr(> z)
Intercept	-0.57355	0.22427	0.010544
failures	-1.56207	0.20437	2.12e-14
absences	-0.04457	0.01286	0.000530
studytime	0.28581	0.08173	0.000471
internetyes	0.54570	0.16742	0.001116

Figure B2.3 Reduced Model Coefficients – Model coefficients and statistics for reduced model with only failures, absences, studytime, and internet (top 4 significant predictors).

	Estimate	Std. Error	Pr(> z)
Intercept	0.03989	0.31957	0.90066
sexMale	-0.05883	0.15397	0.70241
failures	-3.79807	0.59032	1.24e-10
absences	-0.055326	0.015002	0.000226

internetyes	0.71241	0.18027	7.75e-05
activitiesyes	0.33791	0.14508	0.01985
goout	-0.20391	0.06490	0.00168
studytime	0.26373	0.08865	0.00293
Dalc	-0.26128	0.09007	0.00372
Walc	-0.007452	0.079125	0.924963

Figure B2.4 Reduced Model with no Influential Points Coefficients – Model coefficients and statistics for a reduced model using the same predictors as the full model.

	Estimate	Std. Error	Pr(> t)
Intercept	10.8316	0.3677	< 2e-16
subjectportuguese	2.1488	0.4633	3.97e-06
Dalc	-0.2811	0.2128	0.187
Subjectportuguese: Dalc	-0.4340	0.2662	0.103

Figure B2.5 Alcohol vs. Subjects Model Coefficients – Coefficients and statistics for a model looking at the interaction between the two subjects (math and Portuguese) and weekday alcohol consumption.

	Estimate	Std. Error	Pr(> t)
Intercept	10.61299	0.41196	< 2e-16
absences	-0.01171	0.01917	0.541541
Dalc	-0.47860	0.13607	0.000455
Sexmale	0.27751	0.25262	0.272231
studytime	0.69817	0.14620	2.05e-06

Figure B2.6 Absences vs. Final Grade Model Coefficients – Coefficients and statistics for a reduced model looking only at absences, Dalc, sex, and studytime.

	Estimate	Std. Error	Pr(> z)
Intercept	1.04691	0.39788	0.00851
failures	-1.54364	0.31022	6.49e-07
internetyes	0.38671	0.28212	0.1705
activitiesyes	0.66185	0.22101	0.00275
studytime	0.15891	0.13027	0.22252
goout	-0.08606	0.10603	0.41698
absences	-0.03230	0.02262	0.15323
Dalc	-0.04899	0.13405	0.71477
Walc	-0.25174	0.11342	0.02644

Figure B2.7 Male Logistic Regression Model – Coefficients and statistics for a model looking at only the male responses with the same predictors as the full model.

	Estimate	Std. Error	Pr(> z)
Intercept	-0.007831	0.418374	0.98507
failures	-1.547287	0.273088	1.46e-08
internetyes	0.68143	0.21763	0.00174
activitiesyes	0.07624	0.18179	0.67496
studytime	0.357719	0.113786	0.00167
goout	-0.098193	0.081075	0.22584
absences	-0.035181	0.015098	0.01980
Dalc	-0.227141	0.160931	0.15812
Walc	0.144706	0.100603	0.15032

Figure B2.8 Female Logistic Regression Model – Coefficients and statistics for a model looking at only the female responses with the same predictors as the full model.

B3. Model Comparison

	df	AIC	BIC	Resid Df	Resid Dev	Deviance	Pr(>Chi)
logit_mod	10	1271.291	1320.799	1039	1271.3		
logit_reduced	5	1281.330	1306.084	1034	1251.3	20.039	0.001229

Figure B3.1 Comparing full model to reduced model with top 4 significant predictors – comparing full model with model that uses failures, absences, studytime, and internet with anova of the full and reduced models.

	df	AIC	BIC
logit_mod	10	1271.291	1320.799
logit_noinfl	10	1142.896	1192.171

Figure B3.2 Comparing full model to model without influential observations – comparing full model with model that removed all influential observations.

C. References

Faraway, J. J. (2016). *Extending the linear model with R: Generalized linear, mixed effects and nonparametric regression models* (2nd ed.). Chapman & Hall/CRC.

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